



Semester: July 2025 – November 2025		
Maximum Marks: 50	Examination: End-Semester Examination	Duration: 2 Hrs.
Programme code: 03 Programme: BTech EXTC	Class: TY BTech EXTC	Semester: V (SVU 2023)
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department: EXTC	
Course Code: 216U03C503	Name of the Course: Digital signal Processing	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		
Que. No.	Question Statement	Max. Marks
Q.1	Attempt any two	10
i)	FIR filters are always stable filters. Justify the statement using example	05
ii)	If $x[n] = [3 \ 2 \ -1 \ 0 \ -2 \ 1 \ 4 \ 1 \ 2 \ 4]$ Sketch the decimated signal if $x[n]$ is decimated by factor of 2	05
iii)	Find auto correlation of $x(n) = [-2, 3, 2, 3, -1]$	05
Q.2	Attempt any one	10
i)	a) Find Difference equation of the system and Draw pole zero plot for the discrete time system with transfer function $H(Z) = \frac{1 - 0.5 Z^{-1}}{1 - z^{-1} + \frac{3}{16} z^{-2}}$ b) Why it is desirable to have linear phase filter? What conditions are to be satisfied by the impulse response of an FIR system in order to have a linear phase?	05 05
ii)	Compute 8 – point DFT of the discrete time signal $x(n) = \{1, 2, 1, 2, 0, 2, 1, 2\}$ Using Radix -2 DIT FFT	10
Q.3	Attempt any one	10
i)	Design a IIR filter with following specifications using Bilinear transformation. Assume Butterworth approximation for design Passband Edge frequency = 0.25π Stopband Edge frequency = 0.55π Passband attenuation ≤ 0.5 dB Stopband attenuation ≥ 15 dB	10
ii)	Use the frequency sampling method to design a 9-tap lowpass FIR filter with a cutoff frequency of 0.25π radians/sample.	10
Q.4	Attempt the following	10
	Realise the transfer function in Direct Form II and cascade form $H(Z) = \frac{(1 + 3/4 Z^{-1} + 1/8 Z^{-2})}{(1 - 5/8 Z^{-1} + 1/16 Z^{-2})}$	10



Q.5	Attempt the following	10
A)	If $x[n] = [2 \ 3 \ 4 \ 5]$ find DFT using any method. Using above result and not otherwise find DFT of $y[n] = x[n-1]$	5
	<u>OR</u>	
A)	Compare DITFFT and DIFFFT methods.	5
B)	Explain the advantage of Bilinear Transformation method over impulse Invariant method of IIR filter design.	5